「Thm. Suppose that f is analytic in $\mathbb{C}_\infty$, except for iso. singularities $z_1, \cdots, z_n, \infty$.
Then, $$\mathrm{Res}(f,\infty) + \sum_{k=1}^{n} \mathrm{Res}(f, z_k) = 0.$$

Proof. By the hypothesis, for some $\delta > 0$, a Laurent's representation
$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$
is valid in $\delta < |z| < \infty$. Specifically, put $\delta = \max\{|z_k| \mid k = 1, 2, \cdots, n\}$.
Choose $R > \delta$, and γ be a curve, $\{z \in \mathbb{C} \mid |z| = R\}$, the clockwise direction.
And, for $|z| = R$, the Laurent's representation of f converges uniformly, so that
$$\int_\gamma f(z)\,dz = \sum_{n=-\infty}^{\infty} a_n \cdot \int_\gamma z^n\,dz = -2\pi i \cdot a_{-1}.$$
So, define $\mathrm{Res}(f,\infty) = -a_{-1}$.

Moreover, $f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$ is analytic for $|z| > R$
iff $f(\frac{1}{z}) = \sum_{n=-\infty}^{\infty} a_n z^{-n}$ is analytic for $0 < |z| < \frac{1}{R}$ $\left(R < \left|\frac{1}{z}\right| \Leftrightarrow |z| < \frac{1}{R} \right)$.
So $a_{-1} = \mathrm{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right)$. ⊥ Actually, just definition.

「Example. Let $f(z) = 1/(z-3)\cdot(z^n - 1)$ $(n \in \mathbb{N})$. Evaluate $I = \frac{1}{2\pi i} \int_{|z|=2} f(z)\,dz$.
By the Residue Theorem, $I = \sum_{i=1}^{n} \mathrm{Res}(f, \exp(2\pi i \cdot \frac{k}{n})) = -\mathrm{Res}(f,\infty) - \mathrm{Res}(f,3)$
Since $\mathrm{Res}(f,\infty) = -\mathrm{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right) = -\mathrm{Res}\left(\frac{1}{z^2} \cdot \frac{z^{n+1}}{(1-z)(1-z^n)}, 0\right) = 0,$
because the rational function is analytic at $z = 0$.
Consequently,
$$I = -\mathrm{Res}(f,3) = -\lim_{z\to 3} \frac{1}{z^n - 1} = \frac{1}{1-3^n},$$
because f has simple pole at $z = 3$.